

Style Transfer from Toxic to Neutral Rewriting

Govind Singhal
Joshua Liu

Problem Definition

Text often contains toxic or harmful language

Examples:

- “You’re an idiot”
- “This idea is stupid”
- Aggressive or insulting tone

Toxic language harms online communication in:

- Social media
- Forums
- Comments

Existing tools:

- Detects toxicity
- Do NOT rewrite into constructive language

Project Goal

- Build a system that:
 - Detects toxic language
 - Highlights problematic phrases
 - Rewrites text into neutral, respectful language
- Key requirement
 - Preserve meaning
 - Maintain fluency
 - Reduce toxicity

Proposed Approach

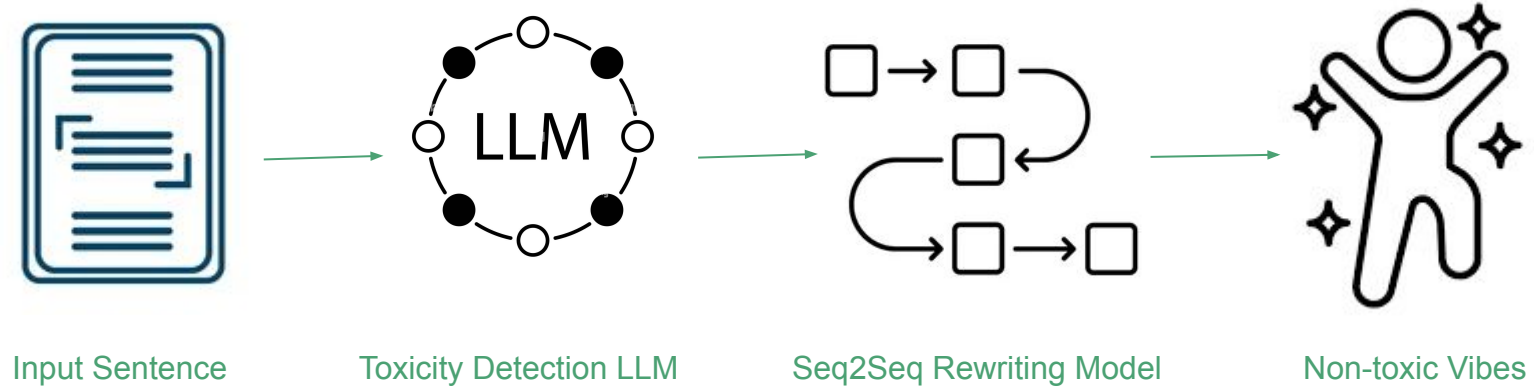
Two-stage pipeline:

1. LLM-based detection
2. Custom Seq2Seq rewriting model

Core Idea:

- LLM handles contextual detection
- Custom model performs controlled rewriting
- Clear separation of tasks between the two models

System Pipeline Architecture



Detection Model

Tasks

- Binary classification
 - Toxic vs non-toxic
- Span detection
 - Identify toxic phrases

Method

- Zero-shot:
 - Model correctly classifies or performs tasks on unseen data without prior training examples
- Few-shot
 - Few examples are provided within a prompt to guide the LLM to better understand tasks
- Structured output

Rewriting Model

Model Type

- Sequence-to-Sequence (Seq2Seq)
 - Encoder-decoder (LSTM/GRU or Transformer)

Task

- Input: toxic sentence
- Output: non-toxic rewrite

Goal

- Preserve meaning
- Remove toxicity
- Maintain fluency

Data

Primary dataset ParaDetox:

- Toxic phrases mapped to non-toxic sentence pairs
- Parallel dataset with more than 10k examples
- <https://github.com/s-nlp/paradetox>

Original	my computer is broken and my phone too!! wtf is this devil sh*t???
Paraphrases	My computer is broken and my phone too! So disappointed! My computer is broken and my phone too, what is this? Both my computer and phone are broken.
Original	sh*t is crazy around here.
Paraphrases	It is crazy around here. Stuff is crazy around here. Something is crazy around here.
Original	delete the page and shut up
Paraphrases	Delete the page and stay silent. please delete the page delete the page
Original	massive and sustained public pressure is the only way to get these b*stards to act.
Paraphrases	Massive and sustained public pressure is the only way to get them to act. massive and sustained preasure is the only way to get these people to act
Original	f*ck you taking credit for some sh*t i wanted to do
Paraphrases	You are taking credit for something I wanted to do You're taking credit fro something i wanted to do.

Evaluation

1. Style Success

- a. Toxicity Score - measured with pre trained toxicity classifier (<https://github.com/unitaryai/detoxify>)
- b. Professional Score - measured with pre trained formality ranker (<https://huggingface.co/s-nlp/roberta-base-formality-ranker>)

2. Meaning Preservation

- a. BERT Score - uses contextual embeddings to compare semantic similarity between generated and reference sentences

3. Fluency

- a. BART Score - evaluate how likely and natural a sentence is

Future Work (Stretch Goals)

1. Deploy an API
 - a. Input: Toxic Sentence
 - b. Output: Non-toxic Sentence
2. Chrome Extension calling the API
3. Slackbot calling the API
4. Discord Bot calling the API